

Negative voltage can also be measured if differential input mode—which is not very commonly used for sensor input—is used. In such case the value in ADC registers is given by:

$$ADC = \frac{(V_{POS} - V_{NEG}) \times GAIN \times 512}{V_{REF}}$$

Circuit description

Fig. 1 shows the interfacing sensor with ADC of AVR microcontroller. At the heart of the circuit is ATmega2560 (IC1). The ATmega2560 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. It features 128kB Flash memory with read-while-write capabilities, 4kB EEPROM, 8kB SRAM, 86 general-purpose input/output (I/O) lines, 32 general-purpose working registers, six flexible timers/counters with compare modes, real time counter (RTC), 10-bit ADC, internal and external interrupts, a serial programmable USART, a programmable watchdog timer with internal oscillator and an SPI serial port.

System clock plays a significant role in the operation of the microcontroller. A 16MHz quartz crystal connected to pins 33 and 34 provides clock pulse to the microcontroller (IC1). Power-on reset is provided by electrolytic capacitor C5 and resistor R2. Switch S1 is used for manual reset. Port pins PB2 (master out, serial data in or MOSI), PB3 (master in, serial out or MISO), and PB1 (serial clock pin or SCK) are used for in-system programming (ISP). IC MAX232 (IC2) is used for serial communication. T_{XDO} and R_{XDO} pins of the microcontroller too are used for serial communication with the help of COM port. Pins R_{XDO} and T_{XDO} of the microcontroller are used to send converted analogue data (through ADC) to the PC through COM port.

ADC channel PF0(ADC0) of microcontroller is used for sensor input. The ADC converts an analogue input voltage to a 10-bit digital value through successive approximation. The converted data can be read from ADC register. It sends ADC conversion

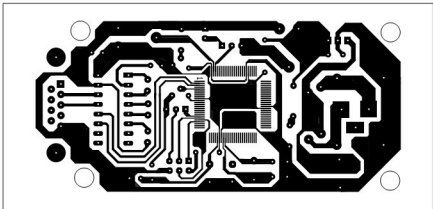


Fig. 2: An actual-size, single-side PCB for the interfacing sensor with ADC of AVR microcontroller

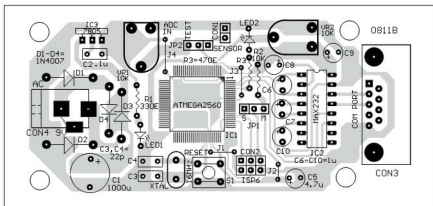


Fig. 3: Component layout for the PCB

results continuously through serial port to the PC at 115.2kbps baud rate. The value received from the ADC pin is reflected in the form of pulse-width modulated (PWM) output at port pin PE3 of the microcontroller, where LED2 is connected to observe the brightness. The ADC contains a sample-and-hold circuit, which ensures that the input voltage to the ADC is held at a constant level during conversion. The ADC has a separate analogue supply voltage pin (AVcc). The reference voltage (V_{REF}) for the ADC is adjusted through preset VR2, which indicates the conversion range for the ADC.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC3). Capacitor C2 bypasses the ripples present in

regulated supply. LED1 acts as the power indicator and resistor R1 limits the current through LED1.

An actual-size, single-side PCB for the circuit of interfacing sensor with ADC of AVR microcontroller is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Assemble the board and check with multimeter. Burn the hex code to microcontroller using suitable programmer through ISP6 connector. Connect the board to the PC's COM port and run serial port interface application.

Serial port interface application window is shown in Fig. 4 for selection of serial COM port. After selection of COM port, set the voltage reference using preset VR2 and apply the test input through preset VR1. Test-input voltage appears along with the voltage